Solve with us 9 - Not Graded

This assignment will not be graded and is only for practice.
Key Points: 1Key Points: 1

- Scalar-valued multivariable functions:Scalar-valued multivariable functions: A scalar-valued multivariable function is a function $f: D \rightarrow R$f : D → R, where DD is a domain in R^n Rn where $n > 1$n > 1.
- Vector-valued multivariable functions:Vector-valued multivariable functions: A vector-valued multivariable function is a function $f: D \rightarrow R^m$ f : D → Rm, where DD is a domain in R^n Rn where $n, m > 1$n, m > 1.

1 point

Which of the following functions are scalar-valued multivariable functions?

- [ ] $f(x, y) = xy + sin(xy)$f(x, y) = xy + sin(xy).
- [ ] $f(x, y, z) = ln(x + y + z)$f(x, y, z) = ln(x + y + z).
- [ ] $f(x, y, z) = (x^2, z, y + xz)$f(x, y, z) = (x2, z, y + xz).
- [ ] $f(x, y) = (xy, 0)$f(x, y) = (xy, 0).
- [ ] $f(x, y) = xy$f(x, y) = xy.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$f(x, y) = xy + sin(xy)$f(x, y) = xy + sin(xy).
$f(x, y, z) = ln(x + y + z)$f(x, y, z) = ln(x + y + z).
$f(x, y) = xy$f(x, y) = xy.

1 point

Which of the following functions are vector-valued multivariable functions?

- [ ] $f(x, y) = xy + sin(xy)$f(x, y) = xy + sin(xy).
- [ ] $f(x, y, z) = ln(x + y + z)$f(x, y, z) = ln(x + y + z).
- [ ] $f(x, y, z) = (x^2, z, y + xz)$f(x, y, z) = (x2, z, y + xz).
- [ ] $f(x, y) = (xy, 0)$f(x, y) = (xy, 0).
- [ ] $f(x, y) = xy$f(x, y) = xy.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$f(x, y, z) = (x^2, z, y + xz)$f(x, y, z) = (x2, z, y + xz).
$f(x, y) = (xy, 0)$f(x, y) = (xy, 0).

Key Points: 2Key Points: 2

- Composition of functions:Composition of functions: Let $D \subseteq R^n$ D ⊆ Rn and $f: D \rightarrow R^m$ f : D → Rm be a multivariable function. Let $g: E \rightarrow R^p$ g : E → Rp be a function on EE where $Range(f) \subseteq E \subseteq R^m$ Range(f) ⊆ E ⊆ Rm.

  We define a multivariable function $g \circ f: D \rightarrow R^p$ g ∘ f : D → Rp called the composition of ff and gg defined as $g \circ f(x\sim) = g(f(x\sim))$g ∘ f(∼x) = g(f(∼x)), for all $x\sim \in D$∼x ∈ D.

1 point

If two functions $f: R^2 \rightarrow R$f : R2 → R and $g: R \rightarrow R$g : R → R are defined as $f(x, y) = (x - y)$ f(x, y) = (x − y) and $g(x) = \sin x$g(x) = sin x, then which of the following functions represents $g \circ f$ g ∘ f?

◯ $\sin x - \sin y$sin x − sin y.

◯ $\sin(x - y)$sin(x − y).

◯ $x \sin x - y \sin y$x sin x − y sin y.

◯ $\sin x\, y$sin xy.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\sin(x - y)$sin(x − y).

1 point

If two functions $f: R^2 \rightarrow R$f : R2 → R and $g: R \rightarrow R^2$g : R → R2 are defined as $f(x, y) = x\, y$ f(x, y) = xy and $g(t) = (e^t, e^{-t})$g(t) = (et, e−t), then which of the following functions represents $g \circ f$ g ∘ f?

◯ $g \circ f(x, y) = e^{x\, y}$ g ∘ f(x, y) = exy.

◯ $g \circ f(x, y) = 1$g ∘ f(x, y) = 1.

◯ $g \circ f(x, y) = (e^{x\, y}, e^{-x\, y})$g ∘ f(x, y) = (exy, e−xy).

◯ $g \circ f(x, y) = (e^x, e^{-x})$g ∘ f(x, y) = (ex, e−x).

◯ $g \circ f(x, y) = (1, 1)$g ∘ f(x, y) = (1, 1).

No, the answer is incorrect.
Score: 0
Accepted Answers:
$g \circ f(x, y) = (e^{x\, y}, e^{-x\, y})$g ∘ f(x, y) = (exy, e−xy).

Key Points: 3Key Points: 3

- Partial derivatives:Partial derivatives: Let $f(x_1, x_2, \ldots, x_n)$f(x1, x2, … , xn) be a function defined on a domain DD in R^n Rn. The partial derivative of ff with respect to x_i xi is the function denoted by $f_{x_i}(x\sim)$fxi(∼x) or $\frac{\partial f}{\partial x_i}(x\sim)$∂xi∂f(∼x) and defined as

$$\frac{\partial f}{\partial x_i}(x\sim) = \lim_{h \to 0} \frac{f(x\sim + he_i) - f(x\sim)}{h}$$∂xi∂f(∼x) = h→0lim hf(∼x+hei)−f(∼x)

where e_i ei is the unit vector $(0, 0, \ldots, 0, 1, 0, \ldots, 0)$(0, 0, … , 0, 1, 0, … , 0) (at i-th coordinate there is 1, and 0 at all the other coordinates).

$$\frac{\partial f}{\partial x_i}(x_1, x_2, \ldots, x_n) = \lim_{h \to 0} \frac{f((x_1, x_2, \ldots, x_n) + h(0, 0, \ldots, 0, 1, 0, \ldots, 0)) - f(x_1, x_2, \ldots, x_n)}{h}$$∂xi∂f(x1, x2 , … , xn) = h→0lim hf((x1,x2,…,xn)+h(0,0,…,0,1,0,…,0))−f(x1,x2,…,xn)

$$\frac{\partial f}{\partial x_i}(x_1, x_2, \ldots, x_n) = \lim_{h \to 0} \frac{f(x_1, x_2, \ldots, x_{i-1}, x_i + h, x_{i+1}, \ldots, x_n) - f(x_1, x_2, \ldots, x_n)}{h}$$∂xi∂f(x1, x2, … , xn) = h→0lim hf(x1,x2,…,xi−1,xi+h,xi+1,…,xn)−f(x1,x2,…,xn)

- To calculate the partial derivative with respect to x_i xi, think of ff only as a function of x_i xi while treating all other variables as constants. Then calculate it as the derivative of a function of one variable.

1 point

Consider the function $f: R^2 \to R$f : R2 → R such that $f(x, y) = x^2 + y^2$ f(x, y) = x2 + y2. Which of the following function will represent f_x fx or $\frac{\partial f}{\partial x}$∂x∂f?

○ $2x + 2y$2x + 2y.

○ $2x$2x.

○ $2x + y^2$ 2x + y2.

○ $x + y^2$ x + y2.

No, the answer is incorrect.

Score: 0

Feedback:

Treat y as constant.

Accepted Answers:

$2x$2x.

1 point

Consider the function $f: D \to R$f : D → R where $D \subset R^2$ D ⊂ R2, defined as:

$$f(x, y) = \begin{cases} \frac{x^2 y}{x^3 + y^3} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$f(x, y) = {x3+y3x2y0if (x, y) = (0, 0)if (x, y) = (0, 0)

What will be the partial derivative of ff with respect to xx at the point $(0, 0)$(0, 0)?

○ 11.

○ 00.

○ 22

○ Cannot be determined.

No, the answer is incorrect.
Score: 0
Feedback:

Accepted Answers:
00.

Key Points: 4Key Points: 4

- Directional derivative:Directional derivative: The rate of change of $f(x_1, x_2, \ldots, x_n)$f(x1, x2, … , xn) in the direction of the unit vector $u = (u_1, u_2, \ldots, u_n)$u = (u1, u2, … , un) is called the directional derivative and is denoted by $D_u f(x_1, x_2, \ldots, x_n)$Duf(x1, x2, … , xn). The definition of the directional derivative is:

$$D_u f(x_1, x_2, \ldots, x_n) = \lim_{h \to 0} \frac{f(x_1 + u_1 h, x_2 + u_2 h, \ldots, x_n + u_n h) - f(x_1, x_2, \ldots, x_n)}{h}$$Duf(x1, x2, … , xn) = $\lim_{h \to 0}$ hf(x1+u1h,x2+u2h,…,xn+unh)−f(x1,x2,…,xn)

- To calculate the directional derivative of a function ff in a particular direction, we have to first find the unit vector uu in that direction and then compute $D_u f$Duf.
- The partial derivatives are special cases of the directional derivative where $u = e_i$u = ei where e_i eiis i i-th vector of the standard ordered basis of R^n Rn.

1 point
Consider the function $f: R^2 \to R$f : R2 → R, such that $f(x, y) = x + y$f(x, y) = x + y. Which of the following options will denote the directional derivative of ff in the direction of the vector v = (1, 1)v = (1, 1)?

[Hint: The given vector vv is not a unit vector. Convert vv to a unit vector $u = \frac{v}{||v||}$u = ||v||v.]

- 2
- $x + y$x + y
- $\sqrt{2}$ 2
- $\sqrt{2}(x + y)$ 2 (x + y)

No, the answer is incorrect.
Score: 0
Feedback:

Accepted Answers:
$\sqrt{2}$ 2

Key Points: 5Key Points: 5

- Limit of a scalar valued multivariable function:Limit of a scalar valued multivariable function: If there exists some real number LL such that $f(x_1, x_2, \ldots, x_n)$f(x1, x2, … , xn) approaches LL as $(x_1, x_2, \ldots, x_n)$(x1, x2, … , xn) approaches $(a_1, a_2, \ldots, a_n)$(a1, a2, … , an), then we write

$$\lim_{(x_1, x_2, \ldots, x_n) \to (a_1, a_2, \ldots, a_n)} f(x_1, x_2, \ldots, x_n) = L$$(x1,x2,…,xn)→(a1,a2,…,an)lim f(x1, x2, … , xn) = L

- For functions of one variable $f(x)$f(x), there are only two ways in which xx can approach a point aa, from left or from right. But for multivariable functions, $(a_1, a_2, \ldots, a_n)$(a1, a2, … , an) can be approached along any curve.
- If CC is a curve passing through $(a_1, a_2, \ldots, a_n)$(a1, a2, … , an) , the limit of ff at $(a_1, a_2, \ldots, a_n)$(a1, a2, … , an) along the curve CC exists and equals LL if $f(x_1, x_2, \ldots, x_n)$f(x1, x2, … , xn) approaches LL as $(x_1, x_2, \ldots, x_n)$(x1, x2, … , xn) approaches $(a_1, a_2, \ldots, a_n)$(a1, a2, … , an) where $(x_1, x_2, \ldots, x_n)$(x1, x2, … , xn) belongs to CC.
- The limit of the function at $(a_1, a_2, \ldots, a_n)$(a1, a2, … , an) along the curve CC equals LL for each curve CC passing through $(a_1, a_2, \ldots, a_n)$(a1, a2, … , an) precisely when the limit of ff at $(a_1, a_2, \ldots, a_n)$(a1, a2, … , an) equals LL.
- To show that the limit of ff at $(a_1, a_2, \ldots, a_n)$(a1, a2, … , an) does not exist, one should either produce a curve CC so that the limit of ff at $(a_1, a_2, \ldots, a_n)$(a1, a2, … , an) along CC does not exist or produce two curves so that the limits of ff at $(a_1, a_2, \ldots, a_n)$(a1, a2, … , an) along both curves are not equal.
- Continuity of a function:Continuity of a function: If limit of a function $f: D \to R$f : D → R, where $D \subseteq R^n$ D ⊆ Rn, exists at a point $(a_1, a_2, \ldots, a_n)$(a1, a2, … , an), and the functional value and the limiting value are the same, i.e.,

$$\lim_{(x_1, x_2, \ldots, x_n) \to (a_1, a_2, \ldots, a_n)} f(x_1, x_2, \ldots, x_n) = f(a_1, a_2, \ldots, a_n)$$(x1,x2,…,xn)→(a1,a2,…,an)lim f(x1, x2, … , xn) = f(a1, a2, … , an)

then we say ff is continuous at the point $(a_1, a_2, \ldots, a_n)$(a1, a2, … , an).

Consider the function $f: D \to R$f : D → R where $D \subset R^2$ D ⊂ R2, defined as:

$$f(x, y) = \begin{cases} \frac{x^2 y}{x^3 + y^3} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$f(x, y) = {x3+y3x2y0if (x, y) = (0, 0)if (x, y) = (0, 0)

Answer the following question 8 to 13 using the above information.

If ff approaches LL as (x, y)(x, y) approaches to the origin along the XX-axis, then find the value of LL.

No, the answer is incorrect.

Score: 0

Feedback:

Along the X-axis, y = 0.

Accepted Answers:

(Type: Numeric) 0

1 point

If f f approaches L L as (x, y)(x, y) approaches to the origin along the Y Y -axis, then find the value of L L.

No, the answer is incorrect.

Score: 0

Feedback:

Along the Y -axis, x = 0.

Accepted Answers:

(Type: Numeric) 0

1 point

If f f approaches L L as (x, y)(x, y) approaches to the origin along the straight line $y = x$ y = x, then find the value of L L.

No, the answer is incorrect.

Score: 0

Feedback:

Accepted Answers:

(Type: Numeric) 0.5

1 point

1 point

If f f approaches L L as (x, y)(x, y) approaches to the origin along the straight line $y = 2x$ y = 2x, then find the value of L L.

◯ 00.

◯ $\frac{1}{2}$ 21.

◯ $\frac{2}{9}$ 92.

◯ 11.

No, the answer is incorrect.

Score: 0

Feedback:

Accepted Answers:

$\frac{2}{9}$ 92.

1 point

If f f approaches to L L as (x, y)(x, y) approaches to the origin along the straight line $y = mx$ y = mx, then find the value of L L.

◯ 00.

◯ $\frac{1}{m^2}$ m21.

◯ $\frac{m}{1+m^3}$ 1+m3m.

◯ 11.

No, the answer is incorrect.

Score: 0

Feedback:

Accepted Answers:

$\frac{m}{1+m^3}$1+m3m.

1 point

Choose the set of correct options.

☐ $\lim_{(x,\,y)\,\to\,(0,0)} f(x,\,y)$(x,y)→(0,0)lim f(x, y) does not exist.

☐ $\lim_{(x,\,y)\,\to\,(0,0)} f(x,\,y) = 0$(x,y)→(0,0)lim f(x, y) = 0.

☐ f f is not continuous at the origin.

☐ $\lim_{(x,\,y)\,\to\,(0,0)} f(x,\,y) = \frac{1}{2}$(x,y)→(0,0)lim f(x, y) = 21.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\lim_{(x,\,y)\,\to\,(0,0)} f(x,\,y)$(x,y)→(0,0)lim f(x, y) does not exist.

f f is not continuous at the origin.

1 point

Consider the function $f: D \to R$f : D → R where $D \subset R^2$ D ⊂ R2, defined as:

$$f(x,\,y) = \begin{cases} \frac{y}{x^2+y} & \text{if } (x,\,y) \neq (0,0) \\ 0 & \text{if } (x,\,y) = (0,0) \end{cases}$$f(x, y) = {x2+yy0if (x, y) = (0, 0)if (x, y) = (0, 0)

Choose the set of correct options.

[Hint: Approach the origin using the curves $y = x^2$ y = x2 and $y = 2x^2$ y = 2x2.]

☐ $\lim_{(x,\,y)\,\to\,(0,0)} f(x,\,y)$(x,y)→(0,0)lim f(x, y) does not exist.

☐ $\lim_{(x,\,y)\,\to\,(0,0)} f(x,\,y) = 0$(x,y)→(0,0)lim f(x, y) = 0.

☐ f f is not continuous at the origin.

☐ $\lim_{(x,\,y)\,\to\,(0,0)} f(x,\,y) = 1$(x,y)→(0,0)lim f(x, y) = 1.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\lim_{(x,\,y)\,\to\,(0,0)} f(x,\,y)$(x,y)→(0,0)lim f(x, y) does not exist.

f f is not continuous at the origin.

Key Points: 6Key Points: 6

- Gradient of a function:Gradient of a function: If ff is a scalar valued function defined on a domain DD in R^n Rn, then the gradient ∇f∇f is defined as the vector valued function $(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \ \dots \ , \frac{\partial f}{\partial x_n})$(∂x1∂f ,∂x2∂f,…,∂xn∂f).
- If ∇f∇f exists and is continuous in an open ball around the point $(a_1 , a_2 , \ \dots \ , a_n)$(a1, a2, … , an), then the directional derivative of the function ff at $(a_1 , a_2 , \ \dots \ , a_n)$(a1, a2, … , an) in the direction of a vector νv is given by the dot product of the gradient of the function at $(a_1 , a_2 , \ \dots \ , a_n)$(a1, a2, … , an) with the unit vector in the direction of νv i.e.,

$$D_\nu f(a_1 , a_2 , \ \dots \ , a_n) = \nabla f(a_1 , a_2 , \ \dots \ , a_n).\frac{\nu}{||\nu||}$$Dvf(a1, a2, … , an) = ∇f(a1, a2 , … , an).||v||v

1 point

Find the directional derivative of the function $f: R^2 \ \rightarrow \ R$f : R2 → R such that $f(x, y) = 3x^2 \ - 2x\, y$ f(x, y) = 3x2 − 2xy in the direction of the vector (3, 4)(3, 4).

$2x - \frac{6y}{5}$2x − 56y.

$10x - 6y$10x − 6y.

$10x$10x.

$2x$2x.

$-\frac{6y}{5}$−56y.

No, the answer is incorrect.
Score: 0

Accepted Answers:

$2x - \frac{6y}{5}$2x − 56y.

1 point

Find the directional derivative of the function $f: R^2 \ \rightarrow \ R$f : R2 → R such that $f(x, y) = 3x^2 \ + 2y^2$ f(x, y) = 3x2 + 2y2 at the point (1, 1)(1, 1) along the direction of it's gradient vector $\nabla f(1, 1)$∇f(1, 1). [Hint: First find out the unit vector in the direction of the vector $\nabla f(1, 1)$∇f(1, 1).]

22

$\sqrt{13}$ 13

$2\sqrt{13}$2 13

$2\sqrt{2}$2 2

No, the answer is incorrect.
Score: 0

Accepted Answers:

$2\sqrt{132}$ 13

$\sqrt{}$